Figure 1. 8-Puzzle: search tree up to depth 3

State: 3 × 3 board · **Initial:** 1 2 3 / _ 4 6 / 7 5 8 · **Actions:** the blank moves Up, Down, Left, Right · **Goal:** 1 2 3 / 4 5 6 / 7 8 _

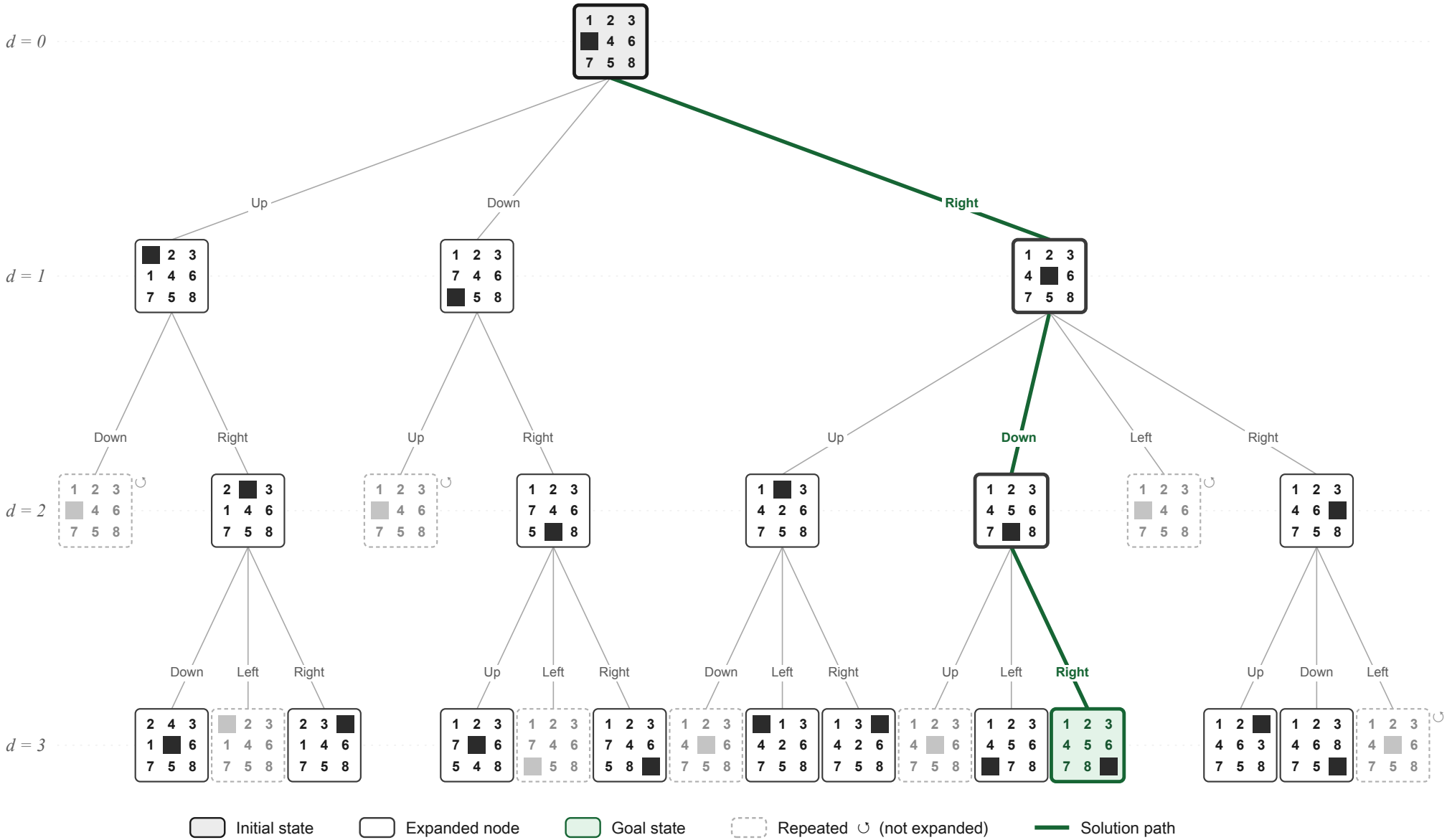


Figure 2. Missionaries and Cannibals: complete search tree

State: (M, C, B) = missionaries and cannibals on the left bank, boat side · **Initial:** (3, 3, L) · **Goal:** (0, 0, R) · **Actions:** the boat carries MM, CC, MC, M or C · **Rule:** M = 0 or M ≥ C on each bank

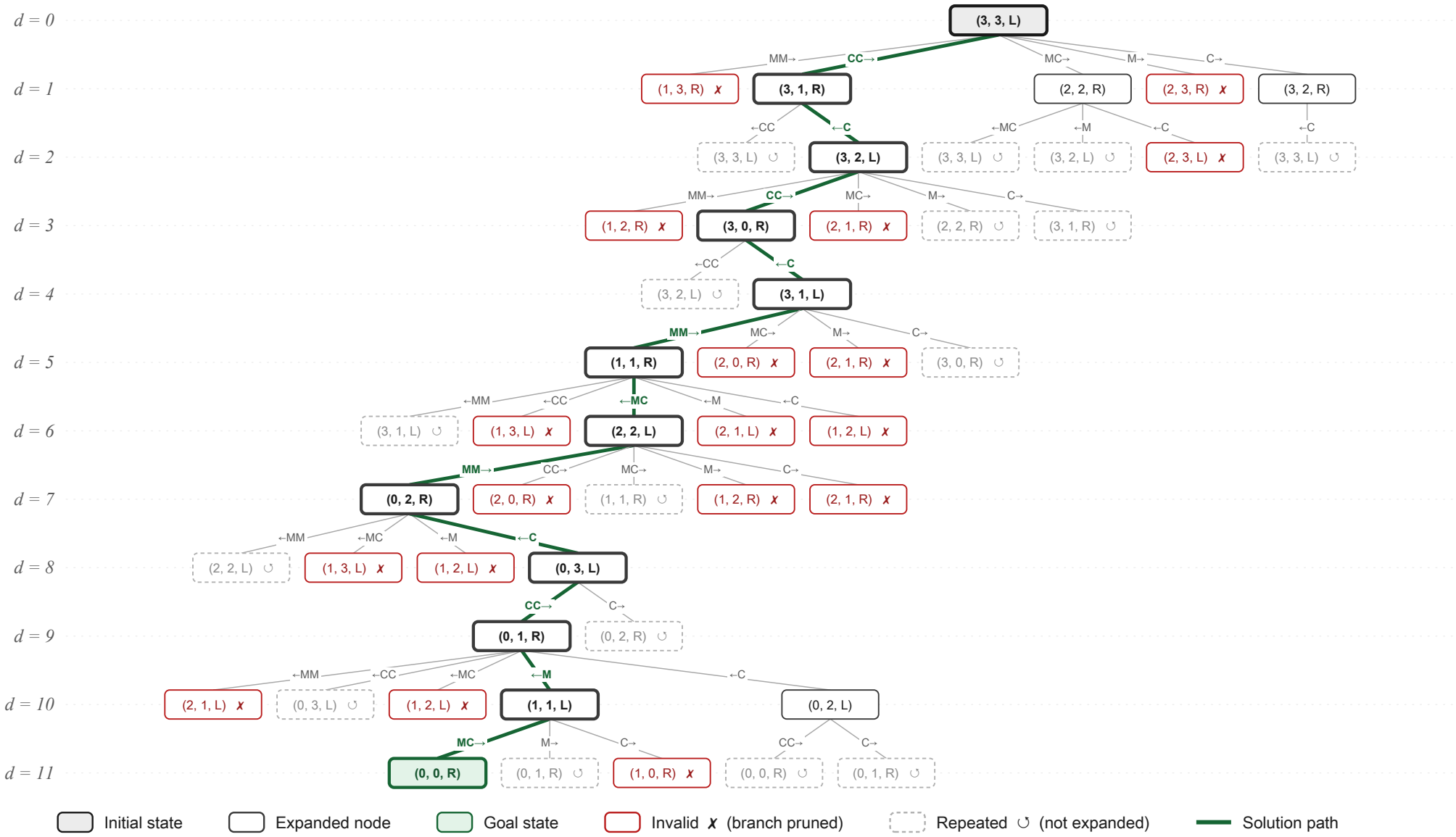


Figure 3. Rotating Workforce Scheduling: first three levels

Instance: rows P1–P3 × 7 days, cyclic · **Shifts:** D day, N night, – off · **Demand:** 1 D and 1 N every day · **R1:** no D right after N · **R2:** max. 4 work days in a row

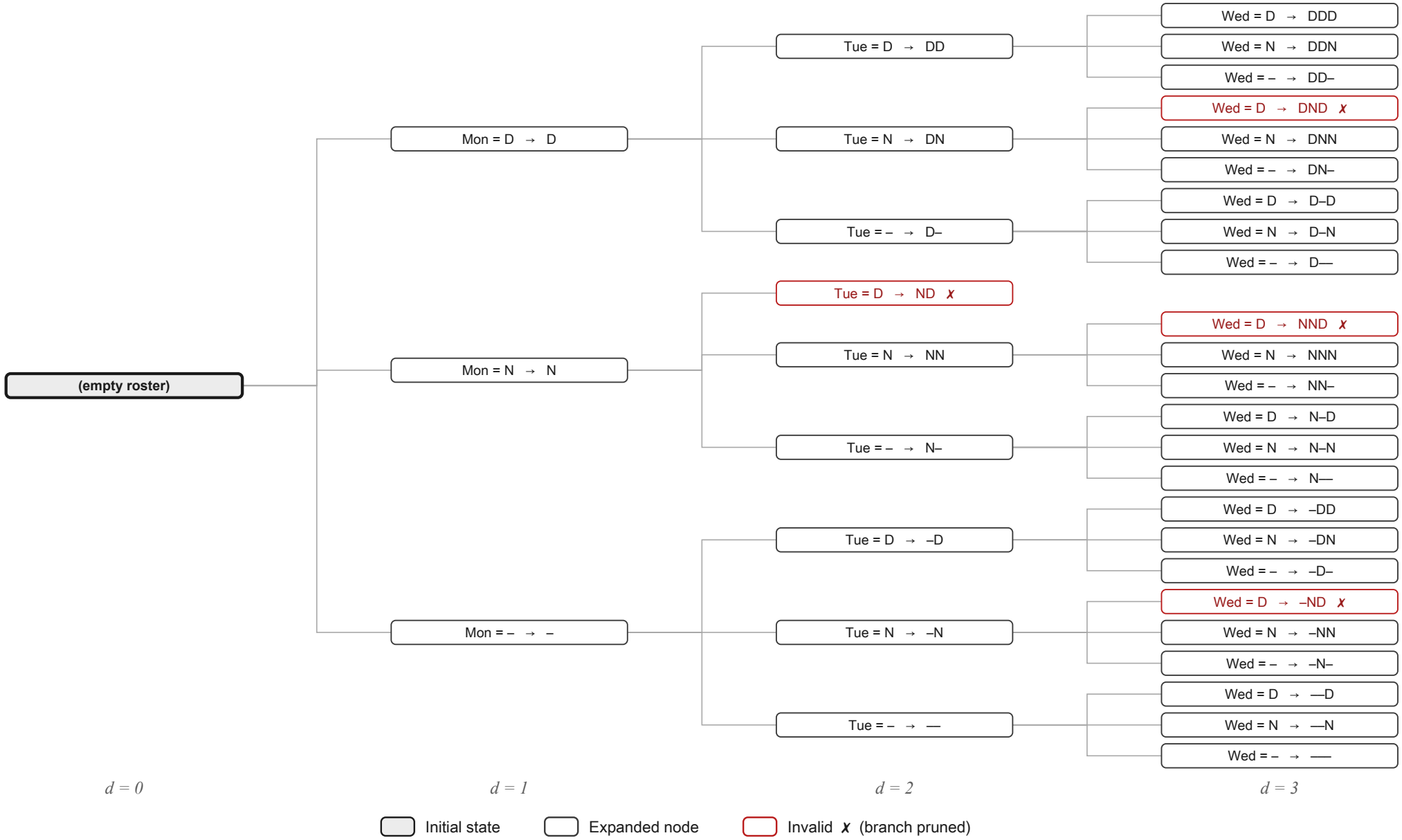


Figure 4a. Rotating Workforce Scheduling: complete DFS path (d = 0 … 11)

Instance: rows P1–P3 × 7 days, cyclic · **Demand:** 1 D and 1 N every day · **R1:** no D right after N · **R2:** max. 4 work days in a row · **DFS order:** D, N, –

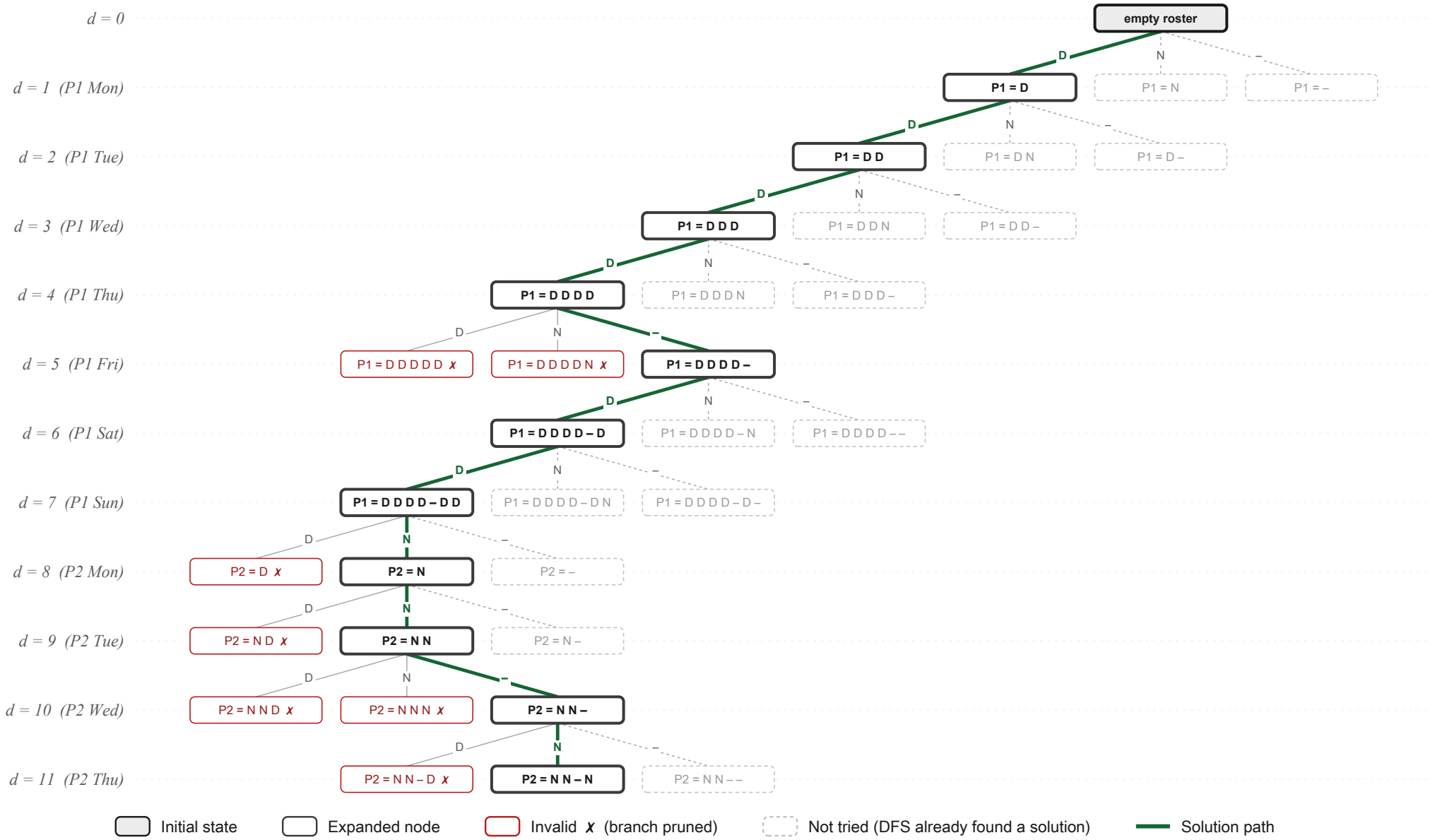


Figure 4b. Rotating Workforce Scheduling: complete DFS path (d = 11 … 21)

Instance: rows P1–P3 × 7 days, cyclic · **Demand:** 1 D and 1 N every day · **R1:** no D right after N · **R2:** max. 4 work days in a row · **DFS order:** D, N, –

